1. BUSINESS PROBLEM

From an initial request to the decision that matters

Our task is to clarify the business decision, not simply produce a list of accounts.

Initial request

"Identify customers showing signs of repayment risk so our operations team can prioritize monitoring."

→

What we need to clarify

Which customers warrant attention?
Should we consider both risk rates and volumes?
How should limited capacity shape priorities?

→

Core business problem

Identify concentrations of observed default risk and allocate monitoring resources with a measurable plan.

DECISION TO SUPPORT

Which groups should receive priority, what support is appropriate, and how will success be assessed?

Context: A historical portfolio case study using UCI Taiwan 2005 data, with 5,000 proactive contacts as a planning capacity assumption.

1. BUSINESS PROBLEM

Turning the management question into a clear analytical approach

Understand capacity, establish priority criteria, then evaluate a practical response.

A. Business need

The monitoring team cannot devote equal attention to every customer.

First, identify where observed default risk is concentrated.

B. Evidence required

Default rate, default cases, segment size, repayment history and changes in status.

Compare both the intensity and volume of risk.

C. Decision required

Define priority segments, suitable support options and a plan to evaluate outcomes.

Select groups for review and design a measurable pilot.

Expected outcome: An evidence based choice and evaluation plan, rather than a ranking built on one metric.

Data scope and classification principles

Use September as the decision point and April through September as the observation window.

PORTFOLIO SIZE

30,000

customers

OBSERVATION WINDOW

6 months

April to September 2005

NEXT MONTH DEFAULTS

6,636

22.12% of customers

Repayment status mapping

−1	PAY_DULY · Payment recorded as due
1 to 9	DELAY · Documented delayed payment
−2 and 0	UNKNOWN · Source definitions unclear

Data owner clarification required

17,496 customers (58.32%) have UNKNOWN as their September status.

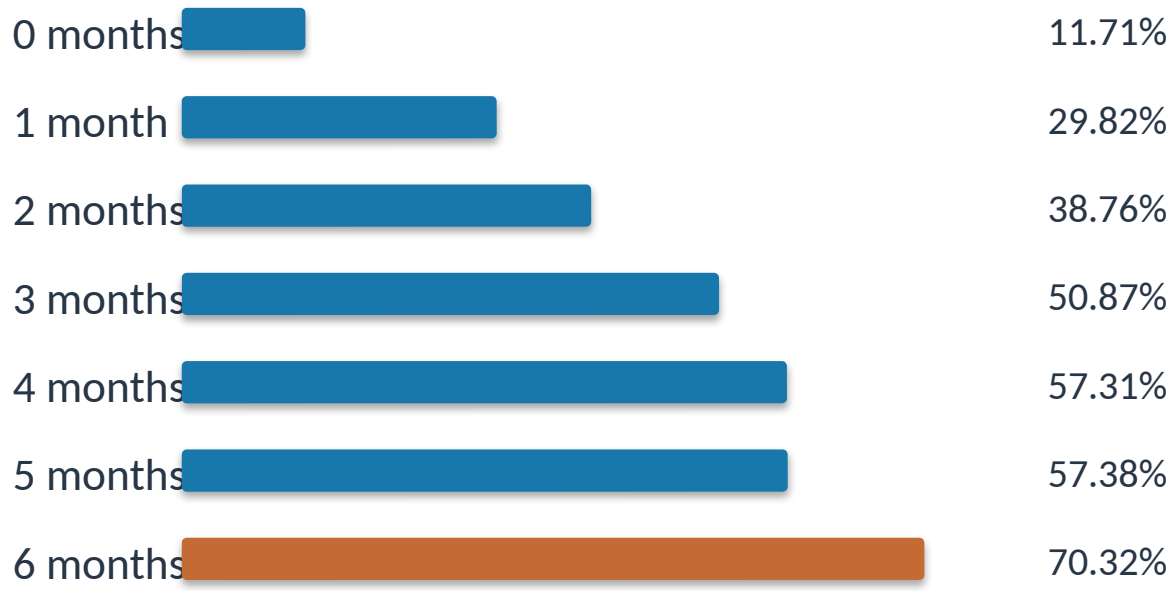
Keep these records separate until their definitions can be confirmed before operational use.

Measurement principle: One default outcome per customer. Portfolio default counts are calculated at customer level.

Repayment history reveals differences in observed default risk

Default rate = customers with a default outcome ÷ all customers in the group × 100%.

Default rate by number of documented DELAY months



Same September status: PAY_DULY

Previous months with DELAY

0 months	13.76%	4,746 customers
1 month	27.97%	572 customers
2 or more	38.32%	368 customers

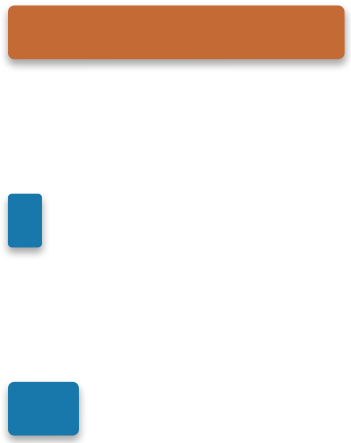
Management insight: A timely September status alone does not capture the differences revealed by earlier repayment history.

3. RISK EVIDENCE

Persistent delayed payment is associated with higher default rates

We examine whether DELAY persists and how the most recent change relates to the next month outcome.

Among monthly transitions starting in DELAY



Default rate by August to September change

DELAY to DELAY	57.78%
DELAY to PAY_DULY	37.78%
PAY_DULY to DELAY	24.66%
PAY_DULY to PAY_DULY	15.19%

Management insight: Persistent DELAY deserves closer review. A single improved status does not erase the earlier history.

Recommended customer segments for risk monitoring

Priorities reflect default rates, default volumes and the number of customers to review.

PRIORITY 1 · DETAILED REVIEW

September DELAY with 2 or more DELAY months

4,449 customers • 2,575 defaults • DR 57.88%

14.83% of customers account for 38.80% of observed defaults.

PRIORITY 2 · EARLY MONITORING

DELAY with 1 month: 2,369 customers

PAY_DULY with 2 or more months: 368 customers

Combined: 2,737 customers • 995 defaults • DR 36.35%

Coverage option	Share of customers	Share of defaults
Priority 1	14.83%	38.80%
Priority 1 and 2	23.95%	53.80%

The two priorities include 7,186 customers. Rollout should reflect capacity and account eligibility.

5. RECOMMENDED ACTIONS

Tailored monitoring and support by customer segment

A tiered approach directs resources toward the needs of each customer group.

PRIORITY 1 September DELAY with 2 or more DELAY months	RECOMMENDED RESPONSE Review account status and eligibility. Prioritize appropriate payment assistance and timely reminders.	INTENDED OUTCOME Limit further deterioration and assess response to support.
PRIORITY 2 DELAY with 1 month; PAY_DULY with 2 or more	RECOMMENDED RESPONSE Monitor the next reporting period and offer light touch reminders where appropriate.	INTENDED OUTCOME Address emerging concerns with minimal disruption.
OTHER CUSTOMERS Groups outside the two priorities	RECOMMENDED RESPONSE Continue regular monitoring and clarify UNKNOWN codes through data governance.	INTENDED OUTCOME Maintain broad oversight and reliable segmentation.

Implementation condition: Confirm UNKNOWN definitions and contact eligibility before applying outreach rules.

5. RECOMMENDED ACTIONS

Test the proposed support before expanding its reach

The objective is to identify both an appropriate customer group and a response that delivers measurable value.

1 CONFIRM ELIGIBILITY

Review account status, contact permissions and available support options. Identify eligible customers.

2 RUN THE PILOT

Use a treatment and comparison group where feasible. Record assignments and actual outreach.

3 MEASURE RESULTS

Compare next month defaults, status changes, costs, successful contacts and customer feedback.

4 MAKE THE DECISION

Expand, refine or stop based on measured outcomes, available capacity and governance requirements.

Success criteria: A credible improvement in outcomes, acceptable costs and a positive customer experience.

Conclusion and approval requested

An evidence based priority, a clear course of action and a measurable decision process.

RECOMMENDATION

Make September DELAY with at least two DELAY months the first segment for detailed monitoring review.

PRIORITY SEGMENT

4,449

14.83% of all customers

DEFAULT RATE

57.88%

2,575 observed defaults

DEFAULT CONTRIBUTION

38.80%

of 6,636 observed defaults

APPROVAL REQUESTED

01 Authorize status code clarification and eligibility review.

02 Approve the pilot design for the priority segment, with defined evaluation measures.

03 Agree on cost and outcome thresholds for a potential wider rollout.